

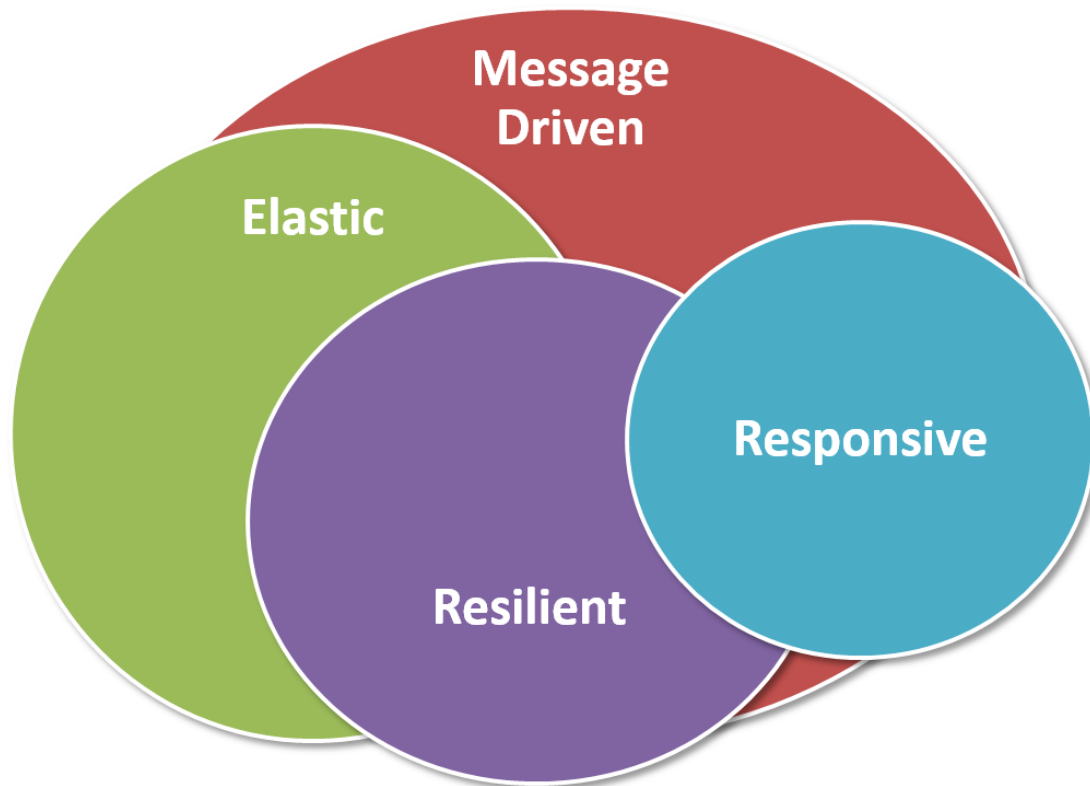
An overview of the event-based microservice architecture

So far, the microservices we've developed are based on REST. We've used REST for both internal (inter-microservice, where one microservice communicates with another microservice in the same system) and external (through the public API) communication. At present, REST fits best for the public API. Are there other alternatives for inter-microservice communication? Is it the best approach to implement the REST for inter-microservice communication? We'll discuss all of this in this section.

You can build microservices that are purely asynchronous. You can build microservice-based systems that can communicate based on events. There's a trade-off between REST and event-based microservices. REST provides synchronous communication, whereas event-based microservices are based on asynchronous communication (asynchronous message passing).

We can use asynchronous communication for inter-microservice communication. Based on the requirements and functionality, we can choose REST or asynchronous message passing. Consider the example of a user placing an order, which makes a very good case for implementing event-based microservices. Upon successful order placement, the inventory service would recalculate the available items, the account service would maintain the transaction, and the correspondence service would send the messages (SMS, emails, and so on) to all involved users, such as the customer and the supplier. In this case, more than one microservice may perform distinct operations (inventory, accounts, messaging, and so on) based on an operation (order placement) performed in one microservice. Now, just imagine if all of these communications were synchronous. Instead, event-based communication, with asynchronous message passing, provides the efficient use of hardware resources, nonblocking, low latency, and high throughput operations.

We can divide the microservice implementations into two groups: REST-based microservices and event-based/message-driven microservices. Event-based microservices are event-based. Event-based microservices are based on the event-based Manifesto (<https://www.event-basedmanifesto.org/>), as shown in the following diagram:



Event-based Manifesto

The event-based Manifesto comprises four principles, which we'll now discuss.

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